

Bachelor Thesis

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**Explainable Machine Learning for Heart Disease Prediction:
A Comparative Analysis of Logistic Regression, Random Forest, and XGBoost**

Negin Hezarjaribi

Supervisor/ Tutor: Prof. Dr. Neil Arvin Bretana

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Abstract

Cardiovascular disease remains one of the leading causes of mortality worldwide, highlighting the need for accurate and transparent prediction methods. This study investigated explainable machine learning for binary heart disease prediction using the processed Cleveland Heart Disease dataset from the UCI Machine Learning Repository. Logistic Regression, Random Forest, and XGBoost were developed and evaluated within a consistent experimental workflow that included data preprocessing, exploratory data analysis, model development, performance evaluation, SHAP-based explainability, and a supplementary evaluation using the Statlog Heart Disease dataset. Model performance was assessed using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix analysis. All three models achieved competitive predictive performance. Logistic Regression obtained the highest ROC-AUC, whereas XGBoost provided the strongest overall balance across the threshold-dependent evaluation metrics and produced the fewest false-negative predictions. SHAP analyses explained both global feature contribution patterns and an individual prediction, with *ca*, *thal*, and *cp* consistently identified as the most influential predictors across the evaluated models. Because all Statlog observations overlapped with records in the processed Cleveland dataset, the supplementary evaluation was not interpreted as independent external validation. Overall, the study demonstrates that combining comparative machine learning evaluation with SHAP-based explainability provides a transparent and reproducible framework for explainable heart disease prediction while emphasising the importance of interpreting predictive performance alongside model behaviour.

Keywords: Heart disease prediction; Machine learning; Explainable artificial intelligence; SHAP; XGBoost

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List of Abbreviations

Abbreviation	Meaning
AI	Artificial Intelligence
F1	F1-score
ML	Machine Learning
ROC	Receiver Operating Characteristic
ROC-AUC	Area Under the Receiver Operating Characteristic Curve
SHAP	SHapley Additive exPlanations
UCI	University of California, Irvine
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting

1 Introduction

1.1 Background

Cardiovascular diseases remain the leading cause of mortality worldwide and continue to impose a substantial burden on healthcare systems. Early identification of cardiovascular disease is therefore important for supporting timely assessment, management, and treatment (World Health Organization, 2025).

In recent years, machine learning has attracted considerable attention as a method for analysing structured clinical data and supporting disease prediction. A wide range of machine learning algorithms has been applied to heart disease prediction. Previous studies have reported encouraging predictive results, indicating that machine learning can identify useful patterns within heart disease datasets (El-Sofany et al., 2024; Mienye & Jere, 2024). However, predictive performance alone provides limited information about how a model arrives at its outputs, which is an important consideration when machine learning is investigated in healthcare-related contexts.

To address this challenge, explainable artificial intelligence techniques have increasingly been incorporated into machine learning workflows. Among these approaches, SHAP (SHapley Additive exPlanations) provides a unified framework for explaining both global feature contribution patterns and individual predictions by assigning feature contribution values to model outputs (Lundberg & Lee, 2017). This improves model transparency and supports a more informed interpretation of model behaviour.

1.2 Problem Statement

Despite the growing number of studies applying machine learning to heart disease prediction, several challenges remain. Many studies place considerable emphasis on predictive performance, while understanding how different models generate their predictions remains an important complementary objective. As a result, a model may achieve strong classification performance while still providing limited insight into the contribution of individual predictor variables and the patterns learned from the data.

Recent studies have increasingly incorporated explainable artificial intelligence into heart disease prediction workflows (El-Sofany et al., 2024; Mesquita & Marques, 2024; Mienye & Jere, 2024). Nevertheless, there remains value in comparing models based on different learning mechanisms within a consistent experimental framework and examining whether their feature contribution patterns are similar or model-dependent. In addition, the independence of evaluation datasets requires careful consideration, because overlap with development data limits the extent to which an evaluation can be interpreted as evidence of external generalisability.

This study addresses these issues by comparing Logistic Regression, Random Forest, and XGBoost within a consistent experimental workflow while integrating global and local SHAP analyses into the evaluation process. The relationship between the Cleveland and Statlog Heart Disease datasets is also examined so that the supplementary evaluation can be interpreted within its appropriate methodological context.

1.3 Research Aim and Objectives

This study aimed to investigate explainable machine learning approaches for binary heart disease prediction by comparing the predictive performance and interpretability of Logistic Regression, Random Forest, and XGBoost. The models were evaluated within a consistent experimental workflow, while SHAP-based explainability was integrated to improve understanding of model behaviour and feature contribution patterns.

To achieve this aim, the study pursued the following objectives:

- To preprocess and explore the processed Cleveland subset of the Heart Disease dataset prior to model development.
- To develop and evaluate Logistic Regression, Random Forest, and XGBoost using a consistent experimental framework.
- To compare the predictive performance of the evaluated models using multiple classification metrics.
- To investigate global model behaviour across the three evaluated models and local model behaviour for the selected XGBoost model using SHAP-based explainability analyses.
- To examine the consistency of feature contribution patterns across the evaluated machine learning models.
- To conduct a supplementary evaluation using the Statlog Heart Disease dataset while accounting for the record-level overlap identified during the study.
- To interpret the findings in relation to previous literature and discuss their implications and limitations.

1.4 Research Questions

This study was guided by the following research questions:

RQ1. How do Logistic Regression, Random Forest, and XGBoost compare in terms of predictive performance for binary heart disease prediction using the processed Cleveland Heart Disease dataset?

RQ2. Which predictor variables contribute most strongly to the predictions of the evaluated machine learning models according to the SHAP analyses?

RQ3. To what extent are the global feature contribution patterns consistent across Logistic Regression, Random Forest, and XGBoost?

RQ4. What predictive performance does the selected XGBoost model achieve on the Statlog Heart Disease dataset, and how should the results be interpreted in light of the identified record-level overlap with the processed Cleveland dataset?

1.5 Scope of the Study

This study focused on the development and evaluation of explainable machine learning models for binary heart disease prediction using the processed Cleveland subset of the Heart Disease dataset obtained from the UCI Machine Learning Repository (Janosi et al., 1989). Three supervised machine learning algorithms—Logistic Regression, Random Forest, and XGBoost—were evaluated within a consistent experimental framework including data preprocessing, exploratory data analysis, model development, performance evaluation, SHAP-based explainability, and supplementary evaluation using the Statlog Heart Disease dataset.

The scope of the study was limited to binary classification following the transformation of the original multi-class target variable. Model performance was assessed using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices. Explainability was investigated through global SHAP analyses for all three models and a local SHAP explanation for the selected XGBoost model.

The study did not seek to develop a new machine learning algorithm or explainability method, nor did it aim to establish clinical or causal relationships between the predictor variables and heart disease. Furthermore, because the record-level comparison demonstrated that all 270 Statlog observations were also present in the processed Cleveland dataset, the Statlog analysis was interpreted as a supplementary evaluation rather than as independent external validation. Consequently, the findings should be interpreted within the methodological and interpretive boundaries established throughout this thesis.

1.6 Structure of the Thesis

This thesis is organised into six chapters. Chapter 1 introduces the research background, problem statement, research objectives, research questions, and scope of the study. Chapter 2 reviews the literature on machine learning for heart disease prediction, explainable artificial intelligence, and SHAP before identifying the research gap. Chapter 3 describes the research methodology, including the datasets, preprocessing procedures, model development, evaluation metrics, SHAP framework, and

supplementary evaluation strategy. Chapter 4 presents the experimental results from the exploratory data analysis, model evaluation, SHAP analyses, and supplementary evaluation. Chapter 5 discusses the findings in relation to previous literature, interprets the SHAP results, considers the practical implications, and outlines the study limitations. Finally, Chapter 6 summarises the study, highlights its main contributions, and presents directions for future research.

2 Literature Review

2.1 Machine Learning in Healthcare

Machine learning (ML), a subfield of artificial intelligence (AI), has become an important technology in modern healthcare by enabling computer systems to identify patterns in complex datasets and support data-driven decision-making. Unlike traditional rule-based systems, machine learning algorithms learn from historical data to recognise relationships between variables and generate predictions for previously unseen cases (Shalev-Shwartz & Ben-David, 2014). With the increasing availability of electronic health records, medical imaging, and clinical databases, machine learning has been applied to disease diagnosis, prognosis, patient risk stratification, treatment recommendation, and many other healthcare applications.

One of the main advantages of machine learning in healthcare is its ability to analyse large numbers of clinical variables simultaneously. Medical data may contain complex and non-linear relationships that are difficult to identify using conventional analytical approaches. Machine learning models can capture these relationships and provide predictive insights that support disease prediction, risk assessment, and clinical decision support, including cardiovascular disease prediction (Srinivas et al., 2010).

Among these applications, cardiovascular disease has received considerable attention because it remains the leading cause of death worldwide (World Health Organization, 2025). Early identification of cardiovascular disease is therefore important for supporting timely clinical decision-making and patient management (World Health Organization, 2025). Machine learning techniques have therefore been widely investigated as tools for analysing routinely collected clinical data and estimating the likelihood of heart disease. Depending on the characteristics of the available data and the research objectives, different algorithms may offer different advantages in terms of predictive performance, computational efficiency, and model interpretability.

Despite their promising predictive capabilities, many machine learning models are often criticised for their limited transparency. Complex models, particularly ensemble methods and other high-performing algorithms, can produce accurate predictions while providing little insight into how these predictions are generated. In healthcare, where decisions may directly influence patient care, understanding the

reasoning behind a model's output is an important consideration. Clinicians and healthcare organisations generally require models that are not only accurate but also interpretable, allowing users to understand which factors contribute to individual predictions and to assess whether these predictions are consistent with existing medical knowledge (Lundberg & Lee, 2017).

The growing demand for transparency has contributed to the development of Explainable Artificial Intelligence (XAI), which aims to improve the interpretability of machine learning models without substantially reducing predictive performance. In healthcare, explainability improves user confidence, facilitates model evaluation, and supports the responsible deployment of artificial intelligence systems (Mesquita & Marques, 2024; Skouteli et al., 2024).

Given these developments, recent research has increasingly focused on improving both predictive accuracy and model transparency. The following sections review previous research on heart disease prediction, explainable artificial intelligence, and SHAP-based model interpretation, providing the theoretical foundation for the methodology adopted in this study.

2.2 Machine Learning for Heart Disease Prediction

Heart disease prediction has become one of the most extensively studied applications of machine learning in healthcare. The increasing availability of publicly accessible clinical datasets has enabled researchers to develop predictive models for identifying individuals at risk of heart disease using demographic information, medical history, laboratory measurements, and physiological characteristics. These predictive models aim to support early risk assessment and complement clinical decision-making rather than replace medical expertise (Srinivas et al., 2010; El-Sofany et al., 2024).

A wide range of machine learning algorithms has been investigated for heart disease prediction. Traditional methods such as Logistic Regression, Decision Trees, Support Vector Machines, and Naïve Bayes have been widely applied to binary heart disease classification. Among these, Logistic Regression remains an important baseline because of its computational efficiency and interpretable coefficients, although its ability to model complex non-linear relationships is limited (Hosmer et al., 2013).

Recent research has increasingly focused on ensemble learning techniques, particularly Random Forest and XGBoost, because of their ability to model complex non-linear relationships among clinical variables. Their strong predictive performance and robustness have made them popular choices for structured clinical datasets and healthcare prediction tasks (Breiman, 2001; Chen & Guestrin, 2016). These characteristics have made Random Forest and XGBoost popular choices in medical prediction studies where datasets contain heterogeneous and potentially non-linear relationships.

Several recent studies have demonstrated the effectiveness of these algorithms in heart disease prediction. For example, El-Sofany et al. (2024) developed machine learning models combined with explainable artificial intelligence techniques and reported that ensemble approaches achieved strong predictive performance while also improving model transparency through SHAP-based explanations. Likewise, Mienye and Jere (2024) proposed an optimised ensemble learning framework incorporating explainable AI and demonstrated that combining high-performing predictive models with model interpretation techniques can improve both predictive capability and the understanding of feature contributions. These studies illustrate the growing interest in balancing predictive accuracy with explainability in healthcare applications.

Despite these advances, differences in dataset selection, preprocessing procedures, model configurations, and evaluation metrics make direct comparisons between published studies challenging. Many studies evaluate a limited set of algorithms or focus primarily on predictive performance without systematically comparing feature contribution patterns across different models. Furthermore, the increasing use of explainability methods has not always been accompanied by a structured comparison of explanations across different model families.

These observations support the need for comparative research that evaluates predictive performance and model explainability within a consistent experimental framework. Accordingly, the present study compares Logistic Regression, Random Forest, and XGBoost using the same cleaned dataset, train-test split, and evaluation metrics. SHAP is then applied to analyse global feature contribution patterns across all three models and provide a local explanation for the selected model. A supplementary evaluation on the Statlog Heart Disease dataset was also included, with its methodological interpretation discussed later in the thesis.

2.3 Explainable Artificial Intelligence (XAI)

The increasing adoption of machine learning in healthcare has highlighted the importance of models that are not only accurate but also transparent and interpretable. Although complex algorithms often achieve high predictive performance, their black-box nature can make it difficult to understand how predictions are generated. This lack of transparency is particularly challenging in healthcare, where clinical decisions require clear and trustworthy justification.

Explainable Artificial Intelligence (XAI) has emerged as a research field that addresses this challenge by providing methods to improve the interpretability of machine learning models. Rather than replacing predictive algorithms, XAI techniques aim to explain how a model reaches a particular prediction and identify the factors that most strongly influence the decision. By increasing model transparency, explainability enables users to better understand model behaviour, identify potential sources of error,

and assess whether the generated predictions are consistent with domain knowledge (Arrieta et al., 2020).

The need for explainability is particularly important in healthcare because predictive models may influence decisions related to diagnosis, risk assessment, and treatment planning. Healthcare professionals are more likely to trust decision support systems when they can understand the reasoning behind the recommendations. Furthermore, explainable models facilitate communication between technical developers and clinical experts, allowing model outputs to be interpreted within an appropriate medical context while supporting model validation and quality assessment (Mesquita & Marques, 2024).

Different explainability techniques have been proposed in recent years. Some models, such as Logistic Regression, are inherently interpretable because their parameters can be directly examined. In contrast, more complex models, including Random Forest and XGBoost, generally require post-hoc explanation methods to improve interpretability. Consequently, a variety of model-specific and model-agnostic explainability techniques have been developed for complex machine learning models (Hosmer et al., 2013).

The integration of explainability techniques into healthcare machine learning reflects a broader shift from focusing solely on predictive accuracy towards developing models that are both accurate and interpretable. Explainability supports transparency, facilitates comparison of feature contributions across models, and improves understanding of model behaviour (Skouteli et al., 2024).

Among the available explainability approaches, SHAP (SHapley Additive exPlanations) has become one of the most widely adopted methods because it provides a theoretically grounded framework for interpreting individual predictions as well as overall model behaviour (Lundberg & Lee, 2017). Owing to its ability to produce both local and global explanations while maintaining consistency across different machine learning algorithms, SHAP has been increasingly applied in healthcare prediction studies. The following section therefore reviews the principles of SHAP and discusses its role in interpreting machine learning models for heart disease prediction.

2.4 SHAP for Model Explainability

Among the various explainability techniques developed in recent years, SHAP (SHapley Additive exPlanations) has become one of the most widely adopted approaches for interpreting machine learning models. Introduced by Lundberg and Lee (2017), SHAP is based on Shapley values from cooperative game theory, where each feature is considered a contributor to the final prediction. The method estimates the contribution of each input feature by fairly distributing the prediction among all features according to their marginal impact on the model output.

A key advantage of SHAP is that it provides consistent explanations across different machine learning algorithms. Unlike model-specific interpretation methods, SHAP can be applied to both inherently interpretable models, such as Logistic Regression, and more complex models, including Random Forest and XGBoost. This flexibility enables researchers to compare feature contributions within a common explainability framework, making SHAP particularly suitable for comparative machine learning studies.

SHAP supports two complementary levels of interpretation: global explainability and local explainability. Global explainability focuses on understanding the overall behaviour of a machine learning model by identifying which features have the greatest influence across the entire dataset. This provides valuable insight into the variables that contribute most strongly to model predictions and allows researchers to compare feature importance patterns between different algorithms. In contrast, local explainability investigates individual predictions by illustrating how specific feature values influence the prediction for a single observation. Local explanations therefore help to understand why a particular prediction was produced rather than describing the behaviour of the model as a whole.

The ability to provide both global and local explanations has contributed to the increasing use of SHAP in healthcare research. Recent studies have applied SHAP to investigate feature importance in heart disease prediction models, demonstrating that explainability techniques can improve transparency while maintaining strong predictive performance (El-Sofany et al., 2024; Mienye & Jere, 2024). By identifying the variables that contribute most strongly to model predictions, SHAP improves understanding of model behaviour and enables comparison of feature contribution patterns across different machine learning algorithms.

Despite its advantages, SHAP also has several limitations. The computational cost may become relatively high for large datasets or highly complex models, particularly when exact Shapley values are estimated. Furthermore, SHAP explains how a model reaches its predictions but does not establish causal relationships between predictor variables and the target outcome. Consequently, SHAP results should be interpreted as explanations of model behaviour rather than evidence of clinical cause-and-effect relationships (Lundberg & Lee, 2017). This distinction is particularly important in healthcare applications, where predictive models are intended to support decision-making rather than replace clinical expertise.

Considering these characteristics, SHAP provides an appropriate explainability framework for the present study. It enables the comparison of feature contribution patterns across Logistic Regression, Random Forest, and XGBoost while also supporting detailed interpretation of individual predictions for the best-performing model. Using a common explainability framework across all models enables a more comprehensive evaluation of model behaviour alongside predictive performance.

2.5 Research Gap

Machine learning-based heart disease prediction has been widely investigated using both traditional classifiers and more complex ensemble methods. Existing studies have demonstrated that models such as Logistic Regression, Random Forest, and XGBoost can provide useful predictive performance when applied to demographic and clinical data. More recent research has also incorporated Explainable Artificial Intelligence techniques, particularly SHAP, to improve the transparency of model predictions (El-Sofany et al., 2024; Mienye & Jere, 2024).

Despite this progress, opportunities remain for focused and reproducible comparative studies. Much of the existing literature emphasises performance metrics such as accuracy, precision, recall, F1-score, and ROC-AUC, while offering less detailed analysis of how different model families arrive at their predictions. Consequently, strong predictive performance is not always accompanied by a systematic assessment of model behaviour and feature contribution patterns.

A further limitation concerns the way explainability is applied. SHAP is frequently used to interpret a single selected model, particularly a high-performing ensemble classifier. Although this provides useful information about that model, it does not necessarily show whether linear and tree-based algorithms identify similar influential features. A structured comparison of SHAP-based feature contribution patterns across Logistic Regression, Random Forest, and XGBoost can therefore provide additional insight into the consistency and differences in model behaviour.

The reproducibility and interpretation of validation results also require careful attention. Performance obtained on an additional dataset may appear to provide evidence of external generalisability; however, this interpretation is only justified when the evaluation dataset is genuinely independent of the data used for model development. Dataset provenance and possible overlap between datasets must therefore be considered before validation results are described as independent evidence of generalisation.

The present study addresses these issues through a consistent experimental framework. Logistic Regression, Random Forest, and XGBoost are developed using the same preprocessing workflow, training and testing subsets, and evaluation measures. Global SHAP explanations are generated for all three models to compare their feature contribution patterns, while a local SHAP explanation is produced for the selected XGBoost model to illustrate an individual prediction.

The study also includes a supplementary evaluation of the selected model on the Statlog Heart Disease dataset. During implementation, a record-level comparison indicated that the Statlog observations were contained within the processed Cleveland dataset. The Statlog evaluation is therefore not presented as independent external validation or as evidence of generalisability to a new patient population. Instead,

it is reported transparently as an additional assessment on a closely related dataset, with the relationship between the two datasets explicitly acknowledged.

By combining comparative model evaluation, cross-model SHAP analysis, local explanation, and transparent reporting of dataset overlap, this study provides a focused examination of explainable machine learning for heart disease prediction. This study contributes to the existing literature through a consistent and reproducible analysis of established machine learning models, their predictive performance, and the feature contribution patterns underlying their predictions, rather than by introducing a new classification algorithm.

2.6 Chapter Summary

This chapter reviewed the theoretical background of machine learning for heart disease prediction, Explainable Artificial Intelligence (XAI), and SHAP-based model explainability. It also identified the research gap addressed in this study, highlighting the importance of evaluating predictive performance and explainability within a consistent experimental framework while considering dataset provenance and validation independence.

Building on this foundation, Chapter 3 presents the research methodology, including the datasets, preprocessing, exploratory data analysis, model development, evaluation metrics, SHAP framework, and supplementary evaluation strategy.

3 Methodology

3.1 Research Design

This study adopted a quantitative experimental research design to investigate the predictive performance and explainability of selected machine learning models for heart disease prediction. The research involved the systematic preparation and analysis of publicly available clinical data, followed by the development and evaluation of three supervised binary classification models.

The Cleveland Heart Disease dataset served as the primary dataset for model development and internal evaluation. After preprocessing, exploratory data analysis was conducted to examine the characteristics of the cleaned dataset before model development.

Three machine learning algorithms were evaluated: Logistic Regression, Random Forest, and XGBoost. Logistic Regression served as the linear baseline, whereas Random Forest and XGBoost represented ensemble tree-based methods capable of modelling complex non-linear relationships. To ensure a fair comparison, all models were trained and evaluated using the same training and testing subsets and the same performance measures.

The cleaned dataset was divided into stratified training and testing subsets. Feature scaling was applied only to Logistic Regression, whereas Random Forest and XGBoost were trained using the original feature values.

Model performance was assessed using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix analysis. Both threshold-dependent and threshold-independent measures were considered because no single metric provides a complete assessment of binary classification performance.

SHAP was applied to investigate model behaviour through global analyses of all three models and a local explanation of the selected XGBoost model. The explainability analysis was limited to interpreting model behaviour and did not support causal or clinical conclusions.

Finally, a supplementary evaluation was performed using the Statlog Heart Disease dataset. Because a record-level comparison showed that the Statlog observations were also present in the processed Cleveland dataset, the analysis was reported as a supplementary evaluation rather than as independent external validation.

The overall workflow included dataset inspection, preprocessing, exploratory data analysis, model development, performance evaluation, SHAP-based explainability, and supplementary evaluation. All analyses were implemented in Python using Google Colab and established libraries for data analysis, machine learning, visualisation, and explainability.

3.2 Datasets

Two publicly available heart disease datasets from the UCI Machine Learning Repository were used in this study. The processed Cleveland Heart Disease dataset served as the primary dataset for exploratory analysis, model development, internal testing, and SHAP-based explainability analysis. The Statlog Heart Disease dataset was subsequently used for the supplementary evaluation of the selected model.

Both datasets contain 13 demographic and clinical predictor variables related to heart disease assessment. However, they differ in the number of observations, target coding, missing-value characteristics, and their role within the experimental design. The following subsections describe each dataset and explain how it was used in the study.

3.2.1 Cleveland Heart Disease Dataset

The primary dataset used in this study was the processed Cleveland subset of the Heart Disease dataset obtained from the UCI Machine Learning Repository. The implementation used the file `processed.cleveland.data`. Although the repository contains heart disease databases from Cleveland,

Hungary, Switzerland, and the VA Long Beach, the processed Cleveland subset has been the most widely used in machine learning research (Janosi et al., 1989). The processed Cleveland dataset contains 303 patient observations, 13 predictor variables, and one target variable. The predictors include demographic information and clinical measurements such as age, sex, chest pain type, resting blood pressure, serum cholesterol, fasting blood sugar, resting electrocardiographic results, maximum heart rate, exercise-induced angina, exercise-induced ST depression, the slope of the peak exercise ST segment, the number of major vessels observed through fluoroscopy, and thalassemia-related test results.

The original target variable, labelled num in the dataset documentation, represents the angiographic diagnosis of heart disease. A value of 0 indicates the absence of heart disease, while non-zero values from 1 to 4 indicate its presence. Because the present study addressed binary classification, the target variable was transformed into two classes representing the absence and presence of heart disease.

Initial inspection identified six incomplete observations: four missing values in the ca variable and two in thal, represented by the symbol '?'. After removing these observations, the final dataset used for analysis contained 297 patient records.

The Cleveland dataset was selected because it provides a structured set of demographic and clinical variables widely used in machine learning research on heart disease prediction. Its widespread use also facilitates comparison with previous studies. Nevertheless, its relatively small sample size and demographic composition were considered when interpreting the findings.

3.2.2 Statlog Heart Disease Dataset

The Statlog Heart Disease dataset was also obtained from the UCI Machine Learning Repository. The implementation used the file heart.dat. The repository describes Statlog Heart as a multivariate classification dataset presented in a slightly different form from the existing Heart Disease databases (UCI Machine Learning Repository, n.d.).

The dataset contains 270 patient observations, 13 predictor variables, and one binary target variable. Its predictor variables correspond closely to those in the processed Cleveland dataset, representing the same demographic and clinical characteristics used for heart disease prediction.

Unlike the processed Cleveland file, the Statlog dataset contains no missing values. Its original target variable is encoded using the values 1 and 2, where 1 represents the absence of heart disease and 2 represents its presence. To align the target encoding with the primary dataset and the trained model, these values were converted to 0 and 1 before evaluation.

The Statlog dataset was initially considered for independent evaluation. However, a record-level comparison showed that all 270 observations were also present in the processed Cleveland dataset.

Consequently, Statlog was retained only for supplementary evaluation while explicitly acknowledging the overlap between the two datasets.

The precise role and interpretation of this supplementary evaluation are described further in Section 3.8.

3.3 Data Preprocessing

Data preprocessing was performed to prepare the Cleveland Heart Disease dataset for machine learning model development. The objective of this stage was to ensure that the dataset was complete, consistently formatted, and suitable for binary classification while preserving the integrity of the original clinical information. All preprocessing procedures were applied before model training and followed a consistent workflow throughout the study.

The first preprocessing step involved identifying incomplete observations within the dataset. In the original file, missing values were represented by the symbol ?, which was converted to standard missing-value indicators (NaN) to enable subsequent processing in Python. Dataset inspection identified four missing values in *ca* and two in *thal*. Because only six observations contained incomplete information, these records were removed and the dataset index was reset. The remaining *ca* and *thal* values were then converted from character strings to numerical data types. This complete-case approach avoided introducing estimated values through imputation and reduced the dataset from 303 to 297 observations.

Following data cleaning, the original target variable was transformed into a binary classification outcome. In the processed Cleveland dataset, the original target values range from 0 to 4, where 0 indicates the absence of heart disease and non-zero values indicate its presence. Since the objective of this study was to develop binary classification models, all non-zero target values were combined into a single positive class. Consequently, the transformed target variable consisted of two classes representing the absence (0) and presence (1) of heart disease.

After target transformation, the predictor variables and the target variable were separated to create the feature matrix and the target vector used for model development. This separation ensured that the machine learning algorithms were trained exclusively on the predictor variables while using the binary target as the classification label.

The cleaned dataset was subsequently divided into training and testing subsets using an 80:20 stratified train-test split. Stratification was applied to preserve the class distribution of the binary target variable in both subsets, while a fixed random state of 42 was specified to improve the reproducibility of the experimental results. Separating the data before model training ensured that model performance was evaluated using previously unseen observations.

Feature scaling was performed only for Logistic Regression because the optimisation procedure of linear models is sensitive to differences in feature magnitude. StandardScaler was fitted exclusively on the training data and subsequently applied to both the training and testing subsets. This approach prevented information from the testing data from influencing the scaling parameters and therefore avoided data leakage. Random Forest and XGBoost were trained using the original feature values because tree-based algorithms are generally unaffected by feature scaling.

3.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted after data preprocessing to obtain an overall understanding of the cleaned dataset before model development. The purpose of this stage was to examine the characteristics of the cleaned dataset and support the subsequent machine learning analysis. The exploratory analysis was descriptive in nature and was not intended to establish predictive relationships between the variables and the target outcome.

The exploratory analysis included an examination of the dataset dimensions, data types, summary statistics, and the distribution of the binary target variable. In addition, the distributions of selected demographic and clinical variables were investigated to assess their overall characteristics and identify potential anomalies that could influence model development.

Continuous variables, including age, resting blood pressure, serum cholesterol, maximum heart rate, and exercise-induced ST depression (oldpeak), were analysed using descriptive summary statistics and visualisations. Measures such as the mean, median, minimum, maximum, and quartiles were examined to describe their central tendency and variability, while the visualisations helped illustrate their distributions and identify potential outliers.

Finally, a Pearson correlation matrix was generated to provide an exploratory overview of pairwise linear associations among the numerically represented predictor variables and the binary target. Because several predictors were binary, categorical, or ordinal variables represented using numerical codes, the resulting coefficients were interpreted cautiously. The matrix was used as an exploratory tool for identifying strong pairwise associations rather than as evidence of causality or direct feature importance. The descriptive findings and corresponding visualisations are presented in Chapter 4.

3.5 Model Development

Three supervised machine learning algorithms were developed and evaluated in this study: Logistic Regression, Random Forest, and XGBoost. These models were selected because they represent different approaches to binary classification while being widely adopted in previous heart disease

prediction research. Evaluating models with different learning mechanisms provides a broader understanding of their predictive performance and behaviour when applied to the same clinical dataset.

3.5.1 Logistic Regression

Logistic Regression was selected as the baseline classification model because it is one of the most widely used supervised learning algorithms for binary classification problems. The model estimates the probability that an observation belongs to one of two classes by modelling the relationship between the predictor variables and the binary outcome using the logistic function. Owing to its relatively simple structure and interpretable coefficients, Logistic Regression provides an appropriate reference model against which more complex machine learning algorithms can be compared (Hosmer et al., 2013).

3.5.2 Random Forest

Random Forest is an ensemble learning algorithm that constructs multiple decision trees using bootstrap sampling and random feature selection (Breiman, 2001). During prediction, the individual tree predictions are aggregated through majority voting to produce the final classification. This ensemble strategy reduces model variance, improves robustness, and decreases the likelihood of overfitting compared with a single decision tree. Random Forest has been widely applied in healthcare prediction because it can effectively model complex relationships among clinical variables while requiring relatively limited data preprocessing.

3.5.3 XGBoost

XGBoost is a gradient boosting algorithm that sequentially builds decision trees, with each new tree aiming to reduce the prediction errors generated by the previous trees. The algorithm incorporates regularisation techniques, efficient optimisation procedures, and parallel computation, making it one of the most widely used machine learning methods for structured tabular data (Chen & Guestrin, 2016). Owing to its ability to capture complex non-linear interactions while maintaining strong predictive performance, XGBoost has been increasingly adopted in healthcare prediction tasks, including heart disease classification.

3.5.4 Model Implementation

All three models were implemented using the Scikit-learn and XGBoost Python libraries within the same experimental environment. To ensure a consistent comparison, the models were trained using the same preprocessed training dataset and evaluated using the same testing dataset.

The models were implemented primarily using the default hyperparameter settings provided by their respective libraries. This approach was intentionally adopted to ensure that the comparison focused on

the inherent characteristics of the three model families rather than on extensive model-specific optimisation. A fixed random state of 42 was specified where applicable to improve the reproducibility of the experimental results. For XGBoost, the evaluation metric was set to log loss during training.

Hyperparameter optimisation was not performed because the primary objective of this study was to compare the predictive behaviour and explainability of Logistic Regression, Random Forest, and XGBoost within a consistent and reproducible experimental framework rather than to maximise the performance of individual models through extensive parameter tuning.

3.6 Model Evaluation

The predictive performance of the developed machine learning models was evaluated using multiple complementary classification metrics. Because no single measure provides a complete assessment of binary classification performance, accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix analysis were used to compare Logistic Regression, Random Forest, and XGBoost (Sokolova & Lapalme, 2009).

Accuracy measures the overall proportion of correctly classified observations but may provide a limited assessment when class distributions are imbalanced. Precision quantifies the proportion of predicted positive observations that are correctly classified, whereas recall (sensitivity) measures the proportion of actual positive observations correctly identified by the model.

The F1-score, calculated as the harmonic mean of precision and recall, provides a balanced measure by considering both false positive and false negative predictions.

The Receiver Operating Characteristic Area Under the Curve (ROC-AUC) evaluates the ability of a model to discriminate between the two classes across all possible classification thresholds. Higher ROC-AUC values indicate stronger discrimination independent of the selected decision threshold (Fawcett, 2006).

Confusion matrices were also generated for each model to summarise the numbers of true positives, true negatives, false positives, and false negatives, providing additional insight into the types of classification errors produced by each algorithm.

All evaluation metrics were calculated using the same testing dataset to ensure a fair and consistent comparison. The comparative evaluation results are presented in Chapter 4.

3.7 SHAP Explainability Framework

To improve the interpretability of the developed machine learning models, SHAP (SHapley Additive exPlanations) was adopted as the primary explainability framework. SHAP is a post-hoc explainability

method based on Shapley values from cooperative game theory that quantifies the contribution of individual features to model predictions (Lundberg & Lee, 2017).

The explainability analysis complemented the quantitative evaluation by investigating how the models generated their predictions and identifying the features that contributed most strongly to the classification outcomes.

Two levels of explainability were incorporated. First, global SHAP analyses were performed separately for Logistic Regression, Random Forest, and XGBoost using the testing dataset. These analyses identified overall feature contribution patterns and enabled comparison of the variables that most strongly influenced each model's predictions. Because SHAP values from different model families may be represented on different output scales, the comparison focused on relative feature rankings rather than raw SHAP magnitudes.

Second, local explainability was performed for the selected XGBoost model to illustrate how individual feature contributions combined to produce a single prediction.

SHAP was applied only after model evaluation and selection to interpret model behaviour rather than to influence model training or optimise predictive performance. The resulting explanations were not interpreted as evidence of causal relationships between clinical variables and heart disease and were not intended to support independent clinical decision-making.

The resulting global and local SHAP analyses are presented in Chapter 4.

3.8 Supplementary Evaluation Strategy

Following the internal evaluation conducted on the Cleveland testing dataset, a supplementary evaluation was performed using the Statlog Heart Disease dataset. The purpose of this stage was to examine the behaviour of the selected model when applied to a second dataset containing the same general predictor structure and binary heart disease classification task.

The Statlog dataset was prepared to match the feature structure and target encoding used during model development. Its original target values of 1 and 2 were converted to 0 and 1, respectively, to represent the absence and presence of heart disease. The predictor columns were then arranged in the same order as those used to train the selected model. No missing-value treatment was required because the Statlog dataset contained complete observations.

The XGBoost model selected during the comparative evaluation was applied directly to the Statlog dataset without retraining, parameter adjustment, or further model optimisation. Predicted classes and positive-class probabilities were generated using the model previously trained on the Cleveland training subset. Performance was then assessed using the same measures applied during the primary

evaluation: accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix analysis. This procedure enabled a direct comparison between performance on the Cleveland testing subset and the Statlog dataset.

The Statlog dataset was initially considered a potential source of independent external validation. However, a record-level comparison performed during implementation indicated that all 270 Statlog observations were also contained within the processed Cleveland dataset used in this study. This finding meant that the independence assumption required for external validation was not satisfied.

Consequently, the Statlog evaluation was reclassified as a supplementary evaluation rather than as independent external validation. Its results were not interpreted as evidence that the model would generalise to a new patient population or an independently collected clinical dataset. Instead, the evaluation was retained to document the observed performance of the selected model on a closely related dataset while transparently acknowledging the overlap between the data sources.

This distinction was important for avoiding an overstated interpretation of the validation results. Strong performance on the Statlog dataset could reflect the close relationship between the two datasets rather than independent model robustness. The supplementary evaluation therefore served as an additional consistency assessment and as a methodological demonstration of the importance of verifying dataset provenance before drawing conclusions about external generalisability.

3.9 Chapter Summary

This chapter presented the methodology adopted in this study, including the research design, datasets, preprocessing, exploratory data analysis, model development, evaluation framework, SHAP explainability, and supplementary evaluation strategy. It also explained why the Statlog analysis was interpreted as a supplementary evaluation rather than as independent external validation.

The methodology established a transparent and reproducible experimental framework for the analyses presented in Chapter 4, which reports the model performance, SHAP explainability results, and supplementary evaluation findings.

4 Results

This chapter presents the results of the exploratory data analysis, comparative model evaluation, SHAP-based explainability analyses, and supplementary evaluation described in Chapter 3. It reports the characteristics of the cleaned Cleveland dataset, compares the performance of Logistic Regression, Random Forest, and XGBoost, presents the SHAP analyses, and concludes with the supplementary evaluation on the Statlog Heart Disease dataset.

4.1 Exploratory Data Analysis Results

Exploratory data analysis was conducted on the cleaned Cleveland dataset before model development to summarise the target distribution, demographic characteristics, selected clinical variables, and pairwise correlations among the predictor variables.

4.1.1 Dataset and Target Characteristics

After removing six incomplete observations, the cleaned dataset contained 297 patient records, 13 predictor variables, and one binary target variable. Following target transformation, 160 observations (53.9%) belonged to the no-heart-disease class and 137 (46.1%) to the heart-disease class. The distribution was sufficiently balanced; therefore, no additional class-balancing technique was applied.

Tab. 1 Distribution of the Binary Target Variable

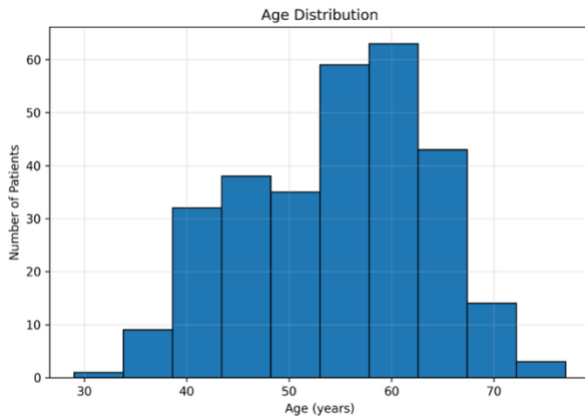
Target class	Number of observations	Percentage
No heart disease	160	53.9%
Heart disease	137	46.1%
Total	297	100.0%

Source: Own representation.

4.1.2 Demographic and Clinical Feature Distributions

The demographic characteristics of the cleaned Cleveland Heart Disease dataset were examined using the age and sex distributions. Patient age ranged from 29 to 77 years, with a mean of 54.5 years and a median of 56 years. The interquartile range extended from 48 to 61 years, indicating that most observations were concentrated within the middle-aged and older adult population (Fig. 1). The dataset contained 201 male (67.7%) and 96 female (32.3%) patients.

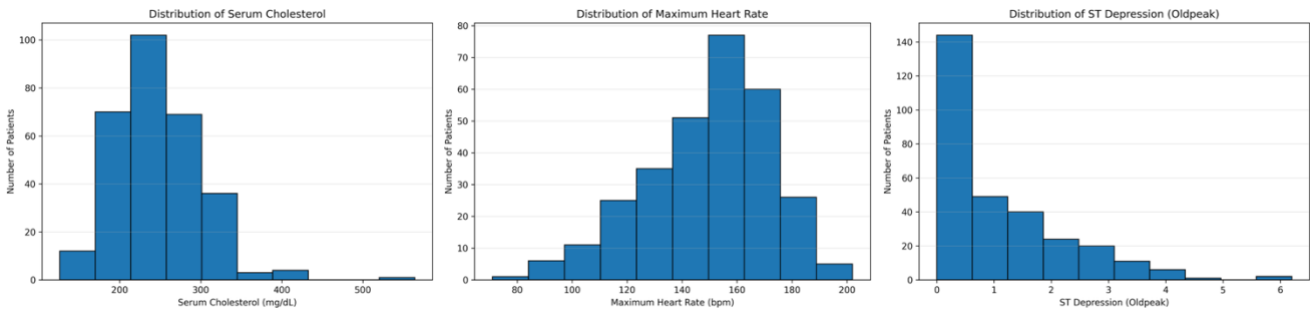
Fig. 1 Age Distribution of the Cleaned Cleveland Dataset



Source: Own representation.

The principal continuous clinical variables were also summarised before model development. Resting blood pressure ranged from 94 to 200 mmHg (mean = 131.7 mmHg). Serum cholesterol ranged from 126 to 564 mg/dL (mean = 247.4 mg/dL), maximum heart rate ranged from 71 to 202 beats per minute (mean = 149.6 bpm), and exercise-induced ST depression (oldpeak) ranged from 0.0 to 6.2 (mean = 1.06). Their distributions are presented in Fig. 2.

Fig. 2 Distributions of Selected Continuous Clinical Variables



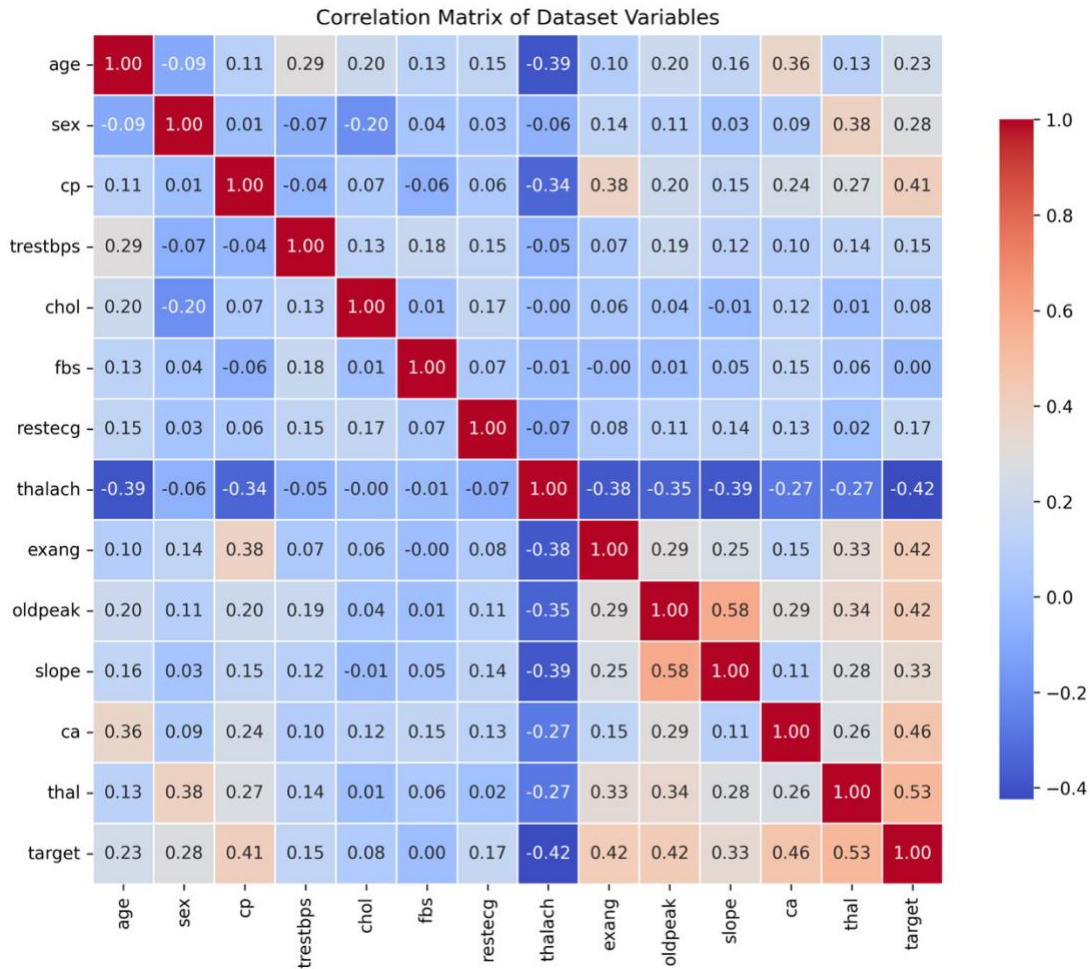
Source: Own representation.

Overall, the descriptive statistics provided an overview of the demographic and clinical characteristics of the cleaned dataset. The relationships among these variables are further explored through the correlation analysis presented in the following section.

4.1.3 Correlation Analysis

To provide an overview of the linear relationships among the variables, a Pearson correlation matrix was generated using the cleaned Cleveland Heart Disease dataset.

Fig. 3 Correlation Matrix of the Cleaned Cleveland Heart Disease Dataset



Source: Own representation.

The correlation matrix showed predominantly weak to moderate pairwise linear associations. The strongest positive association among the predictor variables was observed between oldpeak and slope ($r = 0.58$). With respect to the binary target, the largest positive correlations were observed for thal ($r = 0.53$), ca ($r = 0.46$), exang ($r = 0.42$), oldpeak ($r = 0.42$), and cp ($r = 0.41$), while thalach showed the strongest negative correlation ($r = -0.42$).

No predictor pair exhibited a very strong linear correlation; therefore, no variables were removed based solely on the exploratory correlation matrix. Because several variables were categorical or ordinal features represented by numerical codes, the coefficients were interpreted cautiously and not as measures of feature importance or causal relationships.

4.2 Model Performance Evaluation

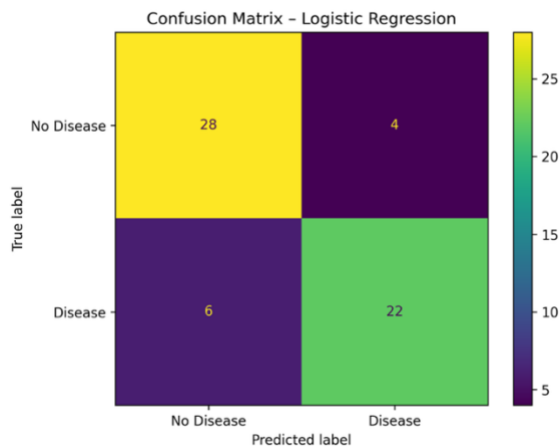
Logistic Regression, Random Forest, and XGBoost were evaluated on the same testing subset of 60 observations using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix analysis. Using the same testing data and evaluation metrics ensured a consistent comparison of model performance.

4.2.1 Logistic Regression

Logistic Regression achieved an accuracy of 0.833, indicating that 83.3% of the observations in the testing subset were classified correctly. The model obtained a precision of 0.846 and a recall of 0.786. Its resulting F1-score was 0.815, while the ROC-AUC score was 0.950.

The confusion matrix in Fig. 4 shows that Logistic Regression correctly classified 28 observations without heart disease and 22 observations with heart disease. Four observations without heart disease were incorrectly classified as positive, while six observations with heart disease were incorrectly classified as negative.

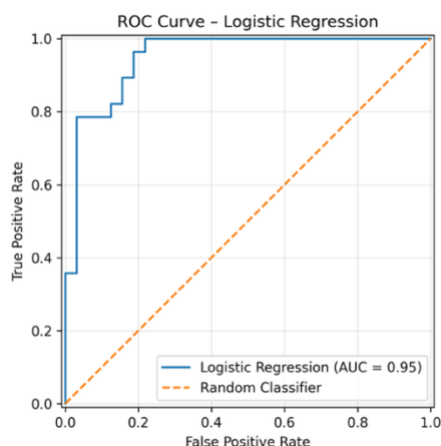
Fig. 4 Confusion Matrix for Logistic Regression



Source: Own representation.

The ROC curve shown in Fig. 5 illustrates the discrimination performance of Logistic Regression across different classification thresholds. The model achieved the highest ROC-AUC value among the three evaluated models (ROC-AUC = 0.950).

Fig. 5 ROC Curve for Logistic Regression



Source: Own representation.

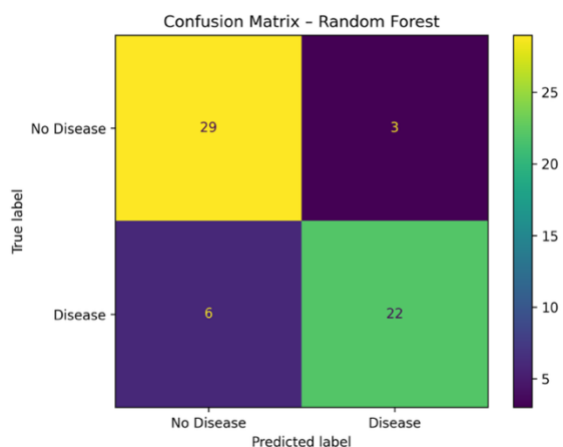
Overall, Logistic Regression provided a strong baseline, achieving the highest ROC-AUC but lower accuracy, precision, and F1-score than Random Forest and XGBoost, while matching the recall of Random Forest.

4.2.2 Random Forest

Random Forest achieved an accuracy of 0.850, indicating that 85.0% of the observations in the testing subset were classified correctly. The model obtained a precision of 0.880 and a recall of 0.786, resulting in an F1-score of 0.830. Its ROC-AUC score was 0.941.

The confusion matrix presented in Fig. 6 shows that Random Forest correctly classified 29 observations without heart disease and 22 observations with heart disease. Three observations without heart disease were incorrectly classified as positive, while six observations with heart disease were incorrectly classified as negative.

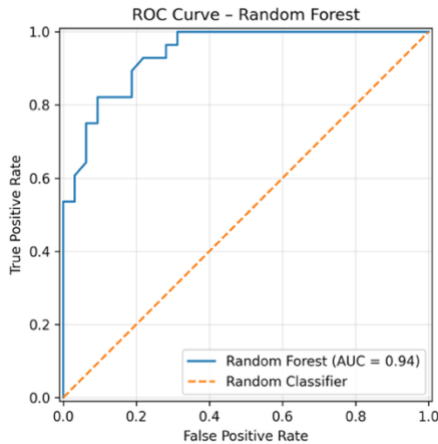
Fig. 6 Confusion Matrix for Random Forest



Source: Own representation.

The ROC curve shown in Fig. 7 illustrates the discrimination performance of Random Forest across different classification thresholds. The model achieved a ROC-AUC value of 0.941.

Fig. 7 ROC Curve for Random Forest



Source: Own representation.

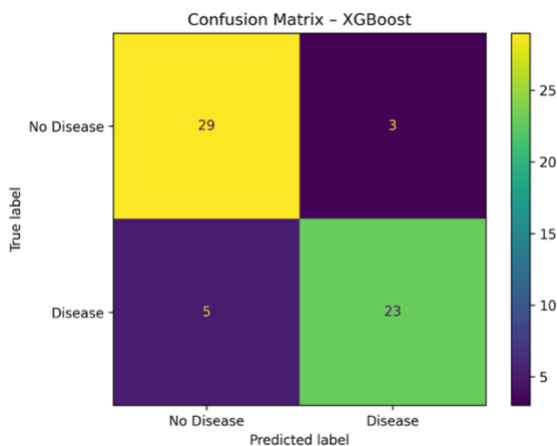
Compared with Logistic Regression, Random Forest achieved slightly higher accuracy, precision, and F1-score while maintaining the same recall. However, its ROC-AUC score was marginally lower than that of Logistic Regression.

4.2.3 XGBoost

XGBoost achieved an accuracy of 0.867, indicating that 86.7% of the observations in the testing subset were classified correctly. The model obtained a precision of 0.885 and a recall of 0.821, resulting in an F1-score of 0.852. Its ROC-AUC score was 0.892.

The confusion matrix presented in Fig. 8 shows that XGBoost correctly classified 29 observations without heart disease and 23 observations with heart disease. Three observations without heart disease were incorrectly classified as positive, while five observations with heart disease were incorrectly classified as negative.

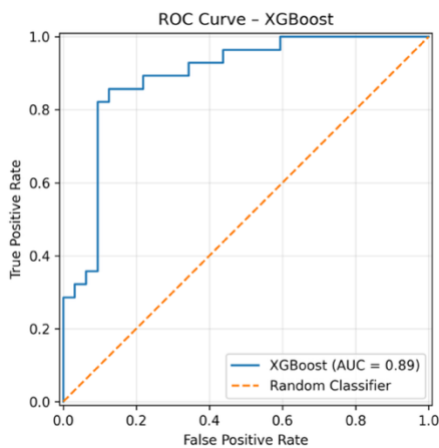
Fig. 8 Confusion Matrix for XGBoost



Source: Own representation.

The ROC curve shown in Fig. 9 illustrates the discrimination performance of XGBoost across different classification thresholds. The model achieved a ROC-AUC value of 0.892.

Fig. 9 ROC Curve for XGBoost



Source: Own representation.

Compared with the other models, XGBoost achieved the highest accuracy, precision, recall, and F1-score. Although its ROC-AUC was lower than that of Logistic Regression and Random Forest, it provided the strongest overall balance across the threshold-dependent evaluation metrics.

4.2.4 Comparative Model Performance and Model Selection

Tab. 2 summarises the comparative performance of Logistic Regression, Random Forest, and XGBoost on the Cleveland testing subset. Although all three models achieved competitive results, their relative performance varied across the evaluation metrics.

Tab. 2 Comparative Performance of the Machine Learning Models

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.833	0.846	0.786	0.815	0.950
Random Forest	0.850	0.880	0.786	0.830	0.941
XGBoost	0.867	0.885	0.821	0.852	0.892

Note. Bold values indicate the highest score obtained for each evaluation metric.

Source: Own representation.

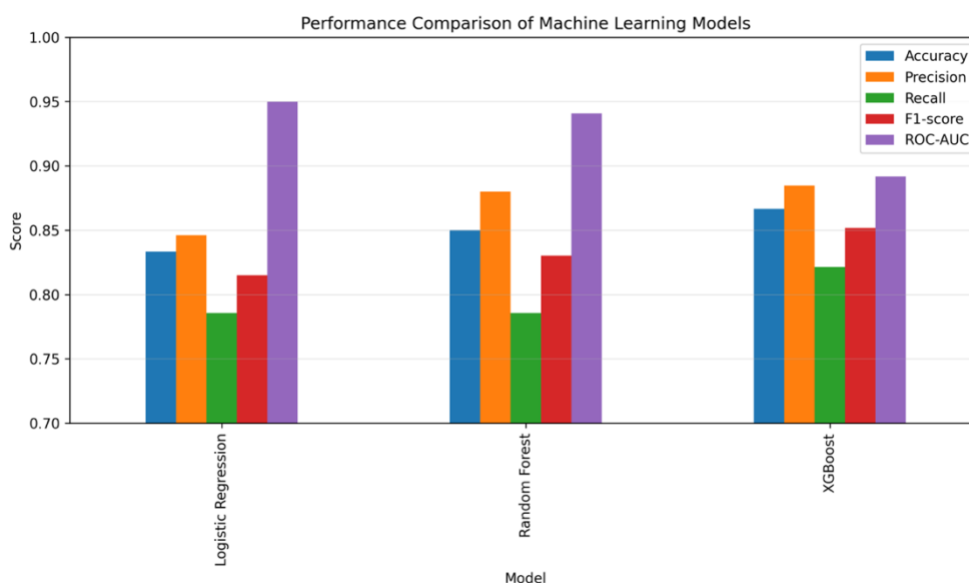
Because the comparison was based on a single testing subset of 60 observations, the differences are reported descriptively and should not be interpreted as evidence of statistically significant superiority.

Among the evaluated models, XGBoost achieved the highest accuracy, precision, recall, and F1-score. At the default classification threshold of 0.5, it correctly classified 52 of the 60 testing observations and produced five false-negative predictions. Logistic Regression and Random Forest each produced six false negatives.

Logistic Regression achieved the highest ROC-AUC score, with a value of 0.950, followed by Random Forest at 0.941 and XGBoost at 0.892. This result indicates that Logistic Regression demonstrated the strongest ranking performance across the full range of classification thresholds, despite obtaining lower threshold-dependent classification scores than XGBoost.

Fig. 10 illustrates the comparative performance of the three models. Differences in accuracy, precision, recall, and F1-score were relatively small, whereas ROC-AUC showed greater variation.

Fig. 10 Performance Comparison of Logistic Regression, Random Forest, and XGBoost



Source: Own representation.

No single model achieved the highest result across every evaluation measure. Model selection was therefore based on the combined pattern of results across the evaluation criteria rather than on a single metric alone. XGBoost was selected for the local SHAP explanation and the supplementary evaluation because it provided the strongest balance of threshold-dependent classification performance, including the highest accuracy, precision, recall, and F1-score, and produced the lowest number of false-negative predictions on the selected testing subset.

This selection does not imply that XGBoost was superior under every evaluation criterion. The higher ROC-AUC obtained by Logistic Regression remained an important comparative finding and was considered when interpreting the overall results. The selection of XGBoost instead reflected the emphasis placed on classification performance at the default decision threshold within the experimental framework of this study.

4.3 Global SHAP Results

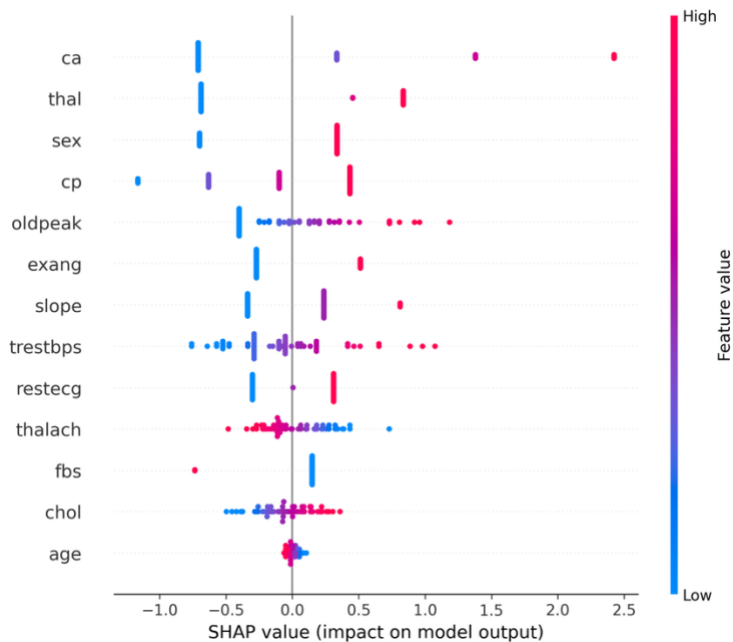
Global SHAP analyses were performed for Logistic Regression, Random Forest, and XGBoost to examine the overall feature contribution patterns identified by each model. For each algorithm, SHAP summary and feature importance plots were generated using the testing dataset. The following sections present the results for each model before comparing their feature contribution patterns across the three algorithms.

4.3.1 Logistic Regression

Global SHAP analysis was performed for the Logistic Regression model using SHAP LinearExplainer (Lundberg & Lee, 2017). Two complementary visualisations were generated: a SHAP summary plot showing the magnitude and direction of feature contributions for individual observations, and a SHAP feature importance plot summarising the mean absolute SHAP value of each predictor.

The global SHAP summary plot is presented in Fig. 11. Predictor variables are ordered according to their overall contribution to the model output, while each point represents an individual observation. The colour scale represents the original feature values.

Fig. 11 Global SHAP Summary Plot for Logistic Regression



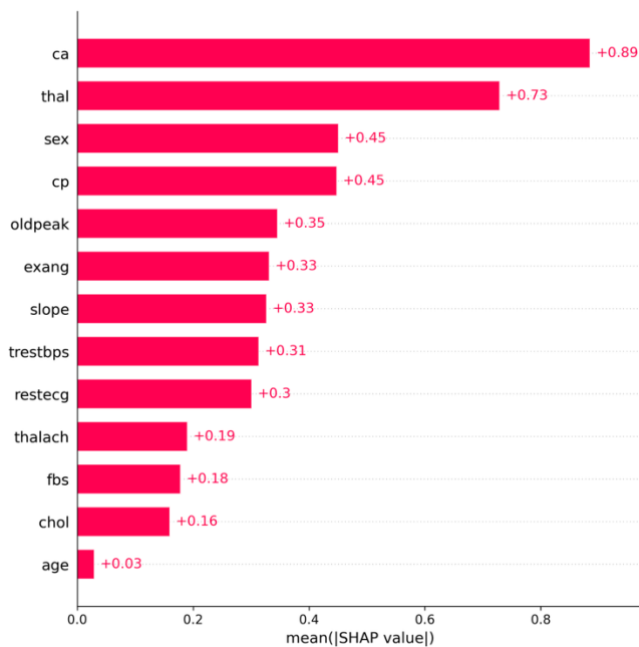
Note. Each point represents one testing observation. Colour indicates the original feature value, while the horizontal position represents the corresponding SHAP value.

Source: Own representation.

The summary plot indicates that *ca* and *thal* contributed most strongly to the Logistic Regression predictions, followed by *sex*, *cp*, *oldpeak*, and *exang*. In contrast, *age* showed the smallest overall contribution among the analysed variables. Higher values of *ca* and observations with higher encoded values of *thal* were generally associated with positive SHAP values within the fitted model. Conversely, higher values of *thalach* were more frequently associated with negative SHAP values, shifting the model output towards the no-heart-disease class.

The global SHAP feature importance plot is presented in Fig. 12. Unlike the summary plot, this visualisation summarises only the average magnitude of feature contributions and does not display their direction.

Fig. 12 Global SHAP Feature Importance for Logistic Regression



Note. Bars represent the mean absolute SHAP value of each predictor across the testing observations.

Source: Own representation.

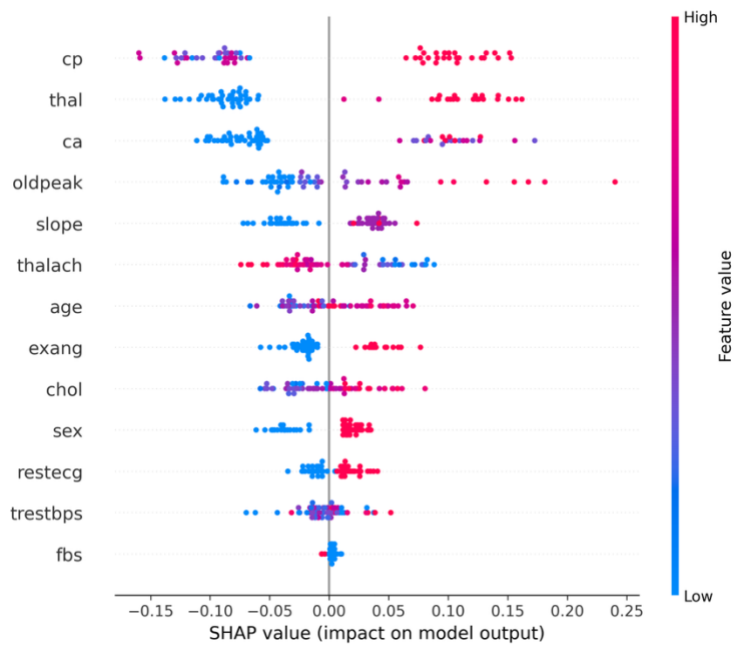
The feature importance plot was consistent with the summary plot. The variables *ca* and *thal* exhibited the largest mean absolute SHAP values, followed by *sex* and *cp*, while *age* showed the smallest overall contribution. Together, these visualisations provide a consistent overview of the relative contribution of each predictor to the Logistic Regression model.

4.3.2 Random Forest

Global SHAP analysis was performed for the Random Forest model using SHAP TreeExplainer (Lundberg & Lee, 2017). SHAP values were calculated from the testing dataset, and two complementary visualisations were generated to examine the overall feature contribution patterns learned by the model.

The global SHAP summary plot is presented in Fig. 13. The predictor variables are ranked according to their overall contribution to the model output, while each point represents an individual observation. The colour scale indicates the original feature values.

Fig. 13 Global SHAP Summary Plot for Random Forest



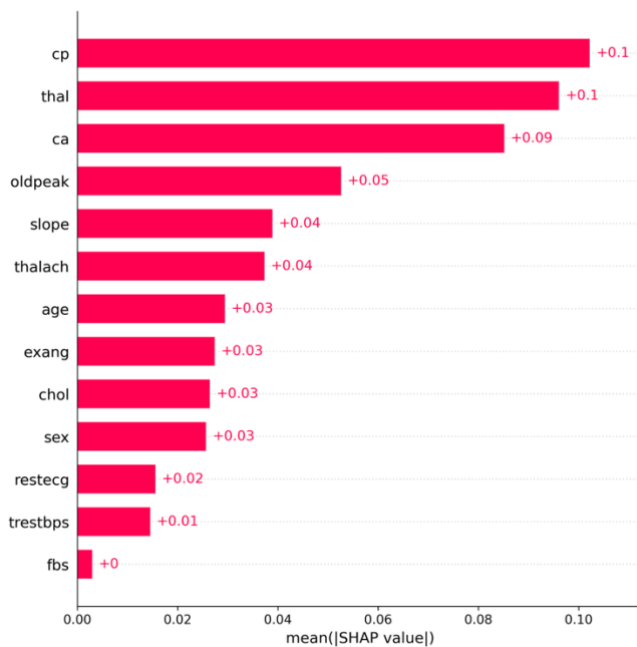
Note. Each point represents one testing observation. Colour indicates the original feature value, while the horizontal position represents the corresponding SHAP value.

Source: Own representation.

The summary plot indicates that cp, thal, and ca contributed most strongly to the Random Forest predictions. These variables exhibited the largest overall SHAP contributions across the testing observations. Higher values of ca and oldpeak, together with observations belonging to specific encoded values of thal and cp, were more frequently associated with positive SHAP values, shifting the model output towards the heart disease class. In contrast, higher values of thalach were more frequently associated with negative SHAP values, shifting the model output towards the no-heart-disease class. The variable fbs exhibited only a limited overall contribution, with SHAP values remaining close to zero for most testing observations.

The global SHAP feature importance plot is presented in Fig. 14. Unlike the summary plot, this visualisation summarises only the average magnitude of feature contributions and does not display their direction.

Fig. 14 Global SHAP Feature Importance for Random Forest



Note. Bars represent the mean absolute SHAP value of each predictor across the testing observations.

Source: Own representation.

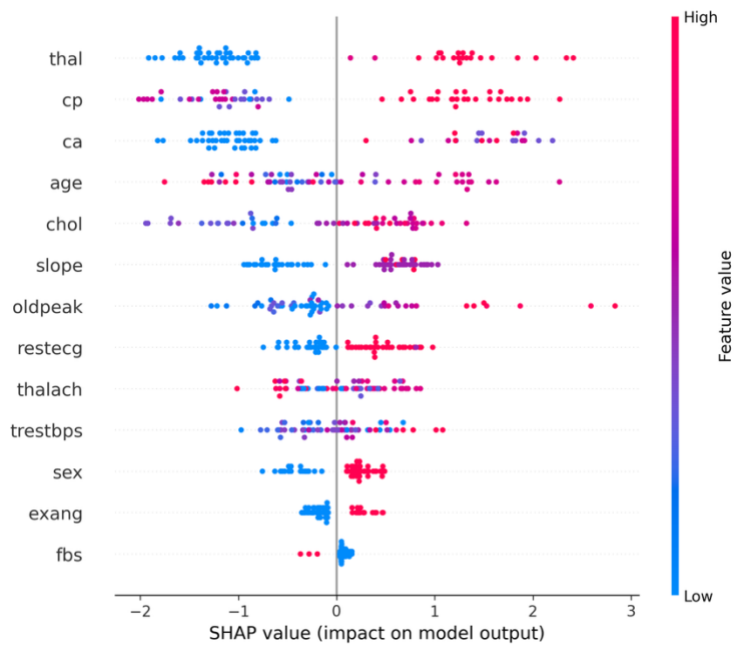
The feature importance plot was consistent with the summary plot. The variables cp, thal, and ca exhibited the largest mean absolute SHAP values, followed by oldpeak and slope. The remaining variables contributed progressively less, with fbs showing the smallest overall contribution. Together, these visualisations provide a consistent overview of the relative contribution of each predictor to the Random Forest model.

4.3.3 XGBoost

Global SHAP analysis was performed for the XGBoost model using SHAP TreeExplainer (Lundberg & Lee, 2017).

The global SHAP summary plot is presented in Fig. 15. Predictor variables are ordered according to their overall contribution to the model output, while each point represents an individual observation. The colour scale indicates the original feature values.

Fig. 15 Global SHAP Summary Plot for XGBoost



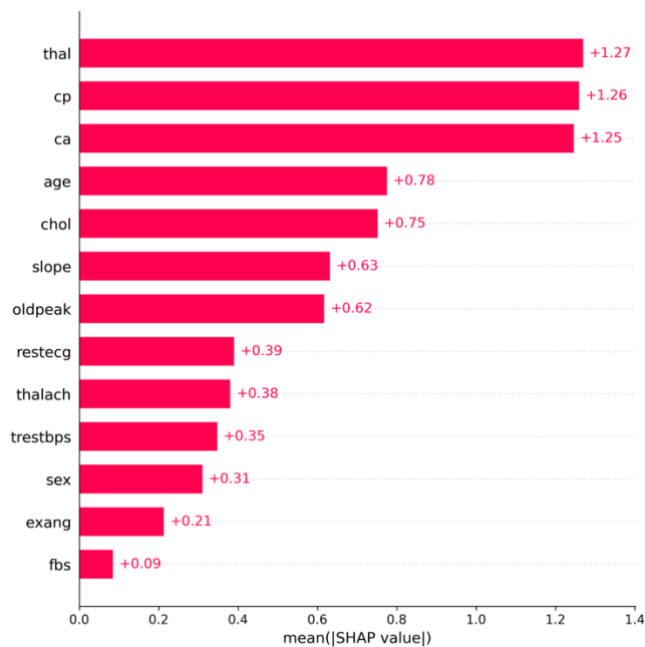
Note. Each point represents one testing observation. Colour indicates the original feature value, while the horizontal position represents the corresponding SHAP value.

Source: Own representation.

The summary plot indicates that thal, cp, and ca contributed most strongly to the XGBoost predictions. Higher values of ca and oldpeak, together with specific encoded values of thal, were generally associated with positive SHAP values, increasing the model output towards the heart disease class. Variables such as age, chol, and slope also contributed to the model predictions, although their overall contributions were considerably smaller than those of the leading predictors.

The global SHAP feature importance plot is presented in Fig. 16. Unlike the summary plot, this visualisation summarises only the average magnitude of feature contributions and does not display their direction.

Fig. 16 Global SHAP Feature Importance for XGBoost



Note. Bars represent the mean absolute SHAP value of each predictor across the testing observations.

Source: Own representation.

The feature importance plot was consistent with the summary plot. The variables `thal`, `cp`, and `ca` exhibited the largest mean absolute SHAP values, while `age`, `chol`, `slope`, and `oldpeak` showed more moderate contributions. The remaining variables contributed relatively little, with `fbs` exhibiting the smallest overall contribution. Together, these visualisations provide a consistent overview of the relative contribution of each predictor to the XGBoost model.

4.4 Comparison of SHAP Results Across Models

The global SHAP analyses presented in the previous sections explained the feature contribution patterns identified by Logistic Regression, Random Forest, and XGBoost. Although the models differ in their underlying learning mechanisms, comparing their highest-ranked features provides insight into whether they relied on similar predictor variables.

To facilitate this comparison, Tab. 3 summarises the five highest-ranked predictor variables identified by each model based on their mean absolute SHAP values. As described in Chapter 3, the comparison focuses on the relative ranking of the features rather than on the raw SHAP magnitudes because SHAP values from different model families are not directly comparable.

Tab. 3 Comparison of the Five Highest-Ranked SHAP Features Across the Three Models

Rank	Logistic Regression	Random Forest	XGBoost
1	ca	cp	thal
2	thal	thal	cp
3	sex	ca	ca
4	cp	oldpeak	age
5	oldpeak	slope	chol

Source: Own representation.

The comparison shows a broadly consistent pattern across the three machine learning models. The variables thal, cp, and ca consistently appeared among the highest-ranked predictors despite differences in the underlying learning algorithms. This agreement indicates that these variables were repeatedly identified as major contributors to model predictions within the cleaned Cleveland Heart Disease dataset.

At the same time, some differences in feature ranking were observed. Logistic Regression assigned relatively greater importance to sex, whereas Random Forest placed greater emphasis on oldpeak and slope. In contrast, XGBoost assigned comparatively higher rankings to age and chol than the other two models. These differences suggest that each learning algorithm captured somewhat different feature contribution patterns while still relying on a common set of leading predictors.

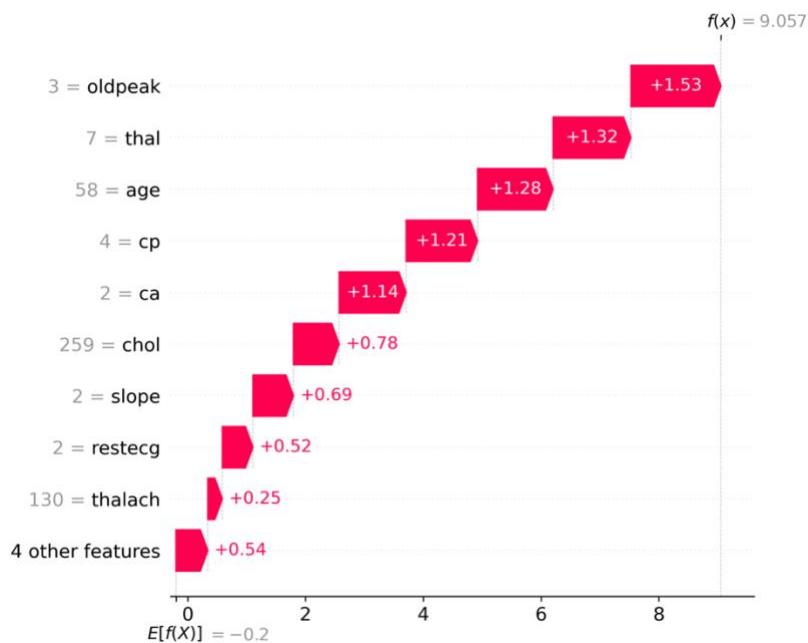
Overall, the cross-model comparison showed broad agreement regarding the variables contributing most strongly to model predictions, while highlighting differences in the relative importance of several secondary features.

4.5 Local SHAP Explanation for XGBoost

While the previous sections examined global feature contribution patterns across the testing dataset, local SHAP analysis explains how the XGBoost model generated the prediction for a single observation. Because XGBoost provided the strongest overall balance across the threshold-dependent evaluation metrics, it was selected for the local SHAP explanation.

To provide a reproducible example, the correctly classified positive observation with the highest predicted probability of heart disease in the testing dataset was selected. This observation corresponded to index 203 in the cleaned Cleveland Heart Disease dataset and was assigned a predicted probability of 0.9999.

Fig. 17 Local SHAP Waterfall Plot for the Selected XGBoost Prediction



Note. The waterfall plot displays SHAP contributions on the raw model output (log-odds) scale. Positive contributions move the prediction towards the heart disease class, while negative contributions move it towards the no-heart-disease class.

Source: Own representation.

The SHAP waterfall plot decomposes the prediction into the contributions of the individual predictor variables, illustrating how the model output moved from its expected baseline value to the final prediction for the selected observation. The plot is displayed on the raw model output (log-odds) scale, while the predicted probability is obtained after applying the probability transformation used by the XGBoost classifier.

For this observation, oldpeak, thal, age, cp, and ca produced the largest positive contributions to the model output. Additional positive contributions were observed for chol, slope, restecg, and thalach, while the remaining predictors collectively made a comparatively smaller contribution. Together, these feature contributions shifted the model output towards a confident prediction of the heart disease class.

The local explanation is broadly consistent with the global SHAP analyses presented earlier in this chapter. Several variables identified among the highest-ranked global contributors, including thal, cp, and ca, also contributed substantially to this individual prediction. This example demonstrates how the global feature contribution patterns learned by the XGBoost model are reflected in a single correctly classified observation.

As the observation was deliberately selected as the highest-confidence correctly classified positive case, the explanation is illustrative and should not be interpreted as representative of the entire dataset.

4.6 Supplementary Evaluation Results

Following the evaluation on the Cleveland testing subset, the selected XGBoost model was further assessed using the Statlog Heart Disease dataset. As discussed in Chapter 3, this analysis is reported as a supplementary evaluation rather than independent external validation because a complete record-level overlap was identified between the Cleveland and Statlog datasets. Consequently, the results are presented as an additional assessment of model performance on a closely related dataset rather than as evidence of independent external generalisability.

4.6.1 Performance on the Statlog Dataset

The trained XGBoost model was evaluated on the Statlog Heart Disease dataset using the same evaluation metrics applied to the Cleveland testing subset. The resulting classification performance is summarised in Tab. 4.

Tab. 4 Performance of XGBoost on the Statlog Heart Disease Dataset

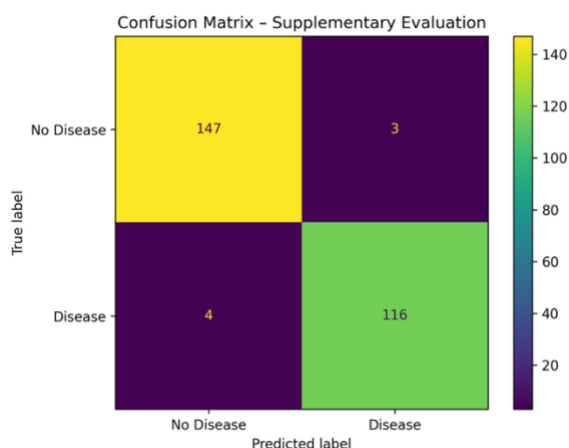
Dataset	Accuracy	Precision	Recall	F1-score	ROC-AUC
Statlog Heart Disease Dataset	0.974	0.975	0.967	0.971	0.984

Source: Own representation.

The supplementary evaluation showed strong predictive performance across all evaluation measures. The model achieved an accuracy of 97.4%, together with high precision, recall, and F1-score values. A ROC-AUC of 0.984 indicates excellent discrimination between the two outcome classes.

The corresponding confusion matrix is presented in Fig. 18.

Fig. 18 Confusion Matrix for the Supplementary Evaluation on the Statlog Heart Disease Dataset

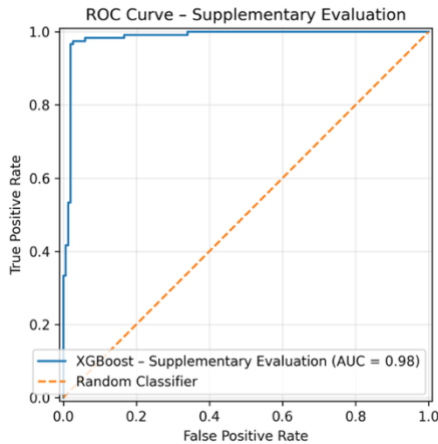


Note. The confusion matrix summarises the classification results obtained on the Statlog Heart Disease dataset. The model correctly classified 147 observations without heart disease and 116 observations with heart disease, while producing three false-positive and four false-negative predictions.

Source: Own representation.

The confusion matrix shows that the model correctly classified the large majority of observations in both outcome classes, producing only seven classification errors across the 270 Statlog observations.

Fig. 19 ROC Curve for the Supplementary Evaluation on the Statlog Heart Disease Dataset



Note. The ROC curve illustrates the trade-off between the true positive rate and false positive rate for the XGBoost model on the Statlog Heart Disease dataset. The area under the curve (ROC-AUC) was 0.984.

Source: Own representation.

Fig. 19 presents the receiver operating characteristic (ROC) curve obtained for the supplementary evaluation. The ROC-AUC value of 0.984 indicates excellent ranking performance across the full range of classification thresholds. These findings should be interpreted in conjunction with the methodological considerations discussed in Chapter 3 regarding the overlap between the Cleveland and Statlog datasets.

4.6.2 Comparison with the Cleveland Test Results

The classification performance on the Statlog Heart Disease dataset was compared with that obtained on the Cleveland testing subset. Tab. 5 summarises the evaluation metrics for both datasets using the selected XGBoost model.

Tab. 5 Comparison of XGBoost Performance on the Cleveland Test Set and Statlog Dataset

Dataset	Accuracy	Precision	Recall	F1-score	ROC-AUC
Cleveland Test Set	0.867	0.885	0.821	0.852	0.892

Statlog Heart Disease Dataset	0.974	0.975	0.967	0.971	0.984
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Source: Own representation.

The supplementary evaluation produced higher values across all reported evaluation measures than those obtained on the Cleveland testing subset. Improvements were observed for accuracy, precision, recall, F1-score, and ROC-AUC.

Although these findings indicate that the selected XGBoost model achieved stronger predictive performance on the Statlog dataset, they should be interpreted with caution. As discussed in Chapter 3, the identified record-level overlap between the Statlog and processed Cleveland datasets means that this evaluation does not represent an independent external validation. Instead, the results provide supplementary evidence that the selected model maintained consistent predictive behaviour on a closely related dataset.

Consequently, the supplementary evaluation provides additional evidence of consistent model performance within this experimental setting but should not be interpreted as evidence of generalisability to an independent patient population.

4.7 Chapter Summary

This chapter presented the results of the exploratory data analysis, comparative model evaluation, SHAP-based explainability analyses, and supplementary evaluation.

The comparative evaluation showed that all three models achieved competitive classification performance, with XGBoost providing the strongest overall balance across the threshold-dependent evaluation measures. The SHAP analyses explained both the global feature contribution patterns and the local behaviour of the selected model. The supplementary evaluation demonstrated strong predictive performance on the Statlog Heart Disease dataset but, because of the identified dataset overlap, was interpreted as supplementary evaluation rather than independent external validation.

These findings provide the basis for the discussion in Chapter 5, where they are interpreted in relation to previous literature, the research objectives, and the limitations of the study.

5 Discussion

This chapter interprets the findings presented in Chapter 4 in relation to the research objectives and previous literature. Rather than restating the experimental results, the discussion examines the performance of the evaluated machine learning models, interprets the SHAP findings, considers the practical implications of the proposed approach, and reflects on the limitations of the study. Together,

these discussions provide a broader understanding of the strengths and limitations of the developed explainable machine learning framework for heart disease prediction.

5.1 Interpretation of Model Performance

The comparative evaluation demonstrated that all three machine learning models achieved competitive predictive performance on the cleaned Cleveland Heart Disease dataset. This finding suggests that the selected predictor variables contained sufficient discriminatory information to distinguish between patients with and without heart disease regardless of the modelling approach. Although Logistic Regression, Random Forest, and XGBoost rely on different learning mechanisms, each model was able to identify meaningful patterns within the dataset and achieve reliable classification performance. These findings indicate that the processed Cleveland dataset provides an appropriate foundation for developing and evaluating machine learning models for binary heart disease prediction.

Among the three evaluated models, XGBoost achieved the strongest overall balance across the threshold-dependent evaluation measures and was therefore selected as the primary model for the explainability analysis and supplementary validation. This finding is consistent with the ability of gradient boosting algorithms to model complex non-linear relationships and feature interactions that are commonly observed in structured tabular datasets (Chen & Guestrin, 2016). Previous studies have similarly reported that XGBoost and related ensemble approaches often achieve strong predictive performance while maintaining robust classification capability across multiple evaluation measures (El-Sofany et al., 2024; Mienye & Jere, 2024). Although the performance differences observed in the present study were relatively small, XGBoost was well suited to the characteristics of the cleaned Cleveland Heart Disease dataset.

Although Logistic Regression achieved the highest ROC-AUC, it was not selected as the primary model because no single evaluation metric was considered sufficient on its own to determine the overall performance of a classifier. ROC-AUC evaluates the ability of a model to discriminate between classes across all possible decision thresholds, whereas accuracy, precision, recall, and F1-score reflect classification performance at the selected decision threshold (Fawcett, 2006). In the present study, XGBoost consistently achieved stronger threshold-dependent classification performance while Logistic Regression demonstrated superior ranking ability. Consequently, the final model was selected on the basis of the combined evidence provided by the evaluation measures rather than the outcome of any single metric.

Overall, the comparative evaluation demonstrated that model selection should be guided by the objectives of the analysis rather than by the optimisation of a single performance metric. Within the

context of the present research, this approach supported the selection of XGBoost as the most appropriate model for the subsequent explainability analysis and supplementary validation.

5.2 Comparison with Previous Literature

The findings of the present study are broadly consistent with previous research on machine learning for heart disease prediction. Consistent with previous studies, all three evaluated models achieved competitive predictive performance, while XGBoost demonstrated the strongest overall balance across the threshold-dependent evaluation measures. These results are consistent with the growing body of evidence indicating that ensemble learning methods can provide highly effective predictive performance for structured clinical datasets while maintaining competitive performance relative to more traditional machine learning approaches (El-Sofany et al., 2024; Mienye & Jere, 2024).

The comparatively strong performance of XGBoost observed in the present study is also supported by recent research in heart disease prediction. El-Sofany et al. (2024) reported that XGBoost achieved high predictive performance while providing interpretable explanations through SHAP, highlighting the suitability of gradient boosting methods for structured clinical data. Similarly, Mienye and Jere (2024) found that ensemble learning approaches combined with explainable artificial intelligence techniques produced robust classification performance and enhanced model interpretability. Although differences in datasets, preprocessing procedures, and experimental designs prevent direct comparison of performance values, the overall agreement with previous studies further supports the suitability of XGBoost for explainable heart disease prediction using structured clinical data.

Although the present findings are broadly consistent with previous research, direct comparison of reported performance values should be undertaken with caution. Differences in dataset characteristics, preprocessing strategies, feature selection procedures, validation strategies, and evaluation settings can all influence the predictive performance of machine learning models. Consequently, variations in reported accuracy or other evaluation measures do not necessarily indicate that one study is methodologically superior to another. Instead, the consistency of the overall performance patterns observed across different studies provides more meaningful support for the suitability of specific machine learning approaches for heart disease prediction.

5.3 Interpretation of SHAP Findings

The SHAP analyses provided additional insight into the behaviour of the evaluated machine learning models beyond their predictive performance. While the classification metrics demonstrated how accurately the models distinguished between the two outcome classes, SHAP explained which predictor variables contributed most strongly to those predictions. Consequently, the explainability analysis

complemented the quantitative performance evaluation by improving model transparency and providing a clearer understanding of how the models generated their predictions (Lundberg & Lee, 2017).

One of the most consistent findings across the SHAP analyses was that several predictor variables, particularly *ca*, *thal*, *cp*, and *oldpeak*, repeatedly appeared among the strongest contributors to the model predictions. Although the relative ranking of these variables varied across the evaluated algorithms, their consistent importance suggests that they contained substantial predictive information within the cleaned Cleveland Heart Disease dataset. Similar feature contribution patterns have been reported in recent explainable machine learning studies, where these variables were likewise identified as influential predictors in heart disease classification models (El-Sofany et al., 2024; Mesquita & Marques, 2024). The overall agreement between the present findings and previous research provides additional support for the interpretation that the explainability analysis captured meaningful feature contribution patterns rather than model-specific artefacts.

Although the three evaluated models consistently identified similar predictor variables as influential, differences were observed in their relative feature rankings. These variations are expected because Logistic Regression, Random Forest, and XGBoost rely on different underlying learning mechanisms. Logistic Regression estimates linear relationships between the predictor variables and the outcome, whereas tree-based models learn hierarchical decision rules and can capture complex non-linear relationships and feature interactions (Breiman, 2001; Chen & Guestrin, 2016). Consequently, differences in SHAP feature rankings should not be interpreted as contradictions but rather as reflections of the distinct ways in which each modelling approach represents the underlying structure of the data.

The local SHAP explanation complemented the global analyses by demonstrating how the selected XGBoost model combined individual feature contributions to generate a single prediction. While the global analyses described overall feature contribution patterns across the testing dataset, the local waterfall plot illustrated how they were reflected in a single correctly classified observation. Several globally important variables, including *thal*, *cp*, and *ca*, also contributed strongly to the selected prediction, while additional features such as *oldpeak* and *age* had substantial case-specific contributions. This illustrates the value of combining global and local explainability, as a variable that is moderately important across the dataset may still have a considerable contribution to a particular prediction (Lundberg & Lee, 2017). However, because the observation was deliberately selected as the highest-confidence correctly classified positive case, the explanation should be regarded as an illustrative example of model behaviour rather than as a representative pattern for all observations.

5.4 Practical Implications

The findings of this study demonstrate that predictive performance and model interpretability can be considered together when developing machine learning models for heart disease prediction. The SHAP analyses showed that, alongside strong classification performance, it is possible to examine how individual predictor variables contribute to model predictions. This combination of predictive modelling and explainability improves model transparency and may support a more informed interpretation of model behaviour in research settings (Lundberg & Lee, 2017).

Beyond the specific findings of the present study, the analytical framework applied in this research may support future investigations of explainable machine learning in healthcare. Combining comparative model evaluation with SHAP-based explainability enables researchers to evaluate both predictive performance and model behaviour within a consistent analytical workflow. This approach may facilitate more transparent comparisons between machine learning models and support future studies using larger, more diverse, and genuinely independent heart disease datasets. Although further validation would be required before any practical clinical implementation could be considered, the methodology presented in this study provides a reproducible framework for future research on explainable machine learning for heart disease prediction.

5.5 Limitations

Several limitations should be considered when interpreting the findings of the present study. First, the analysis was conducted using the Cleveland Heart Disease dataset, which contained a relatively limited number of observations after data cleaning. Although the dataset provided an appropriate basis for the comparative analysis, its limited size may restrict the generalisability of the findings to broader clinical populations. Future studies should therefore evaluate the methodology using larger and more diverse datasets.

A second limitation concerns the supplementary validation performed using the Statlog Heart Disease dataset. As discussed in Chapter 3 and verified through the record-level comparison, all 270 Statlog observations were also present in the processed Cleveland dataset. The Statlog dataset therefore did not represent an independent external validation dataset. Consequently, the supplementary evaluation should be interpreted as an assessment of model performance on a closely related dataset rather than as evidence of generalisability to an independent patient population. Future research should evaluate the methodology using genuinely independent datasets to provide a more rigorous assessment of model generalisability.

Finally, although SHAP improved the interpretability of the evaluated machine learning models, the resulting explanations attribute changes in model output to individual predictor variables rather than

establishing clinical or causal relationships between those variables and heart disease (Lundberg & Lee, 2017). Consequently, the identified feature contribution patterns should be interpreted as explanations of model behaviour within the context of the analysed dataset rather than as evidence of clinical importance. Future research could extend this work by investigating additional explainability techniques, evaluating larger and genuinely independent datasets, and examining the consistency of feature contribution patterns across different clinical populations.

6 Conclusion

This chapter concludes the thesis by summarising the principal findings, highlighting the main contributions of the study, and outlining directions for future research. It brings together the findings from the predictive modelling, SHAP-based explainability, and supplementary evaluation within the methodological boundaries of the study.

6.1 Summary of the Study

This thesis investigated explainable machine learning for binary heart disease prediction using the processed Cleveland Heart Disease dataset. Logistic Regression, Random Forest, and XGBoost were evaluated within a consistent experimental workflow comprising data preprocessing, exploratory data analysis, model development, performance evaluation, SHAP-based explainability, and a supplementary evaluation using the Statlog Heart Disease dataset.

All three models achieved competitive predictive performance, although their strengths differed across the evaluation measures. Logistic Regression achieved the highest ROC-AUC, while XGBoost provided the strongest overall balance across the threshold-dependent metrics and produced the fewest false-negative predictions on the testing subset. Based on the overall evaluation, XGBoost was selected for the local SHAP explanation and the supplementary evaluation.

The SHAP analyses provided both global and local explanations of model behaviour. Across all three models, *ca*, *thal*, and *cp* consistently appeared among the most influential predictors, although their relative importance varied between algorithms. The local SHAP explanation further illustrated how individual feature contributions combined to produce a single prediction.

The supplementary evaluation on the Statlog Heart Disease dataset produced strong predictive performance. However, because all 270 Statlog observations were also present in the processed Cleveland dataset, the analysis was interpreted as a supplementary evaluation rather than independent external validation. Overall, the study demonstrated the value of combining comparative machine learning evaluation with SHAP-based explainability while maintaining a clear distinction between model interpretation and clinical or causal conclusions.

6.2 Research Contributions

This study contributes to the growing body of research on explainable machine learning for heart disease prediction by combining comparative model evaluation with model interpretability within a single analytical framework. Logistic Regression, Random Forest, and XGBoost were evaluated using the same cleaned dataset, preprocessing workflow, experimental protocol, and evaluation measures, enabling a consistent comparison of their predictive performance. This unified approach provided a comprehensive assessment of three widely used machine learning algorithms for binary heart disease prediction.

Beyond comparing predictive performance, this study demonstrated the value of integrating explainability into the machine learning workflow. Global SHAP analyses identified overall feature contribution patterns, while a local SHAP explanation illustrated how individual feature contributions combined to generate a single prediction. Together, these analyses improved the interpretation of model behaviour alongside predictive performance. Furthermore, identifying the record-level overlap between the Cleveland and Statlog datasets highlighted the importance of verifying dataset independence before drawing conclusions about model generalisability. These methodological considerations support more transparent and reproducible explainable machine learning research.

6.3 Future Work

Future research should evaluate the proposed methodology using larger, more diverse, and genuinely independent heart disease datasets to provide a stronger assessment of model generalisability. Repeated cross-validation, independent test cohorts, and formal statistical comparisons could further determine whether observed differences between algorithms are stable rather than specific to a single train-test split.

Future studies could also compare SHAP with other explainability techniques, investigate the consistency of feature contribution patterns across different models, datasets, and populations, and explore hyperparameter optimisation or alternative feature-processing strategies. Any future clinical application would require rigorous external validation, assessment by clinical experts, and careful evaluation of fairness, reliability, and potential risks before implementation in real-world clinical decision support.

Overall, this study demonstrated that combining comparative machine learning evaluation with SHAP-based explainability provides a transparent and reproducible framework for explainable heart disease prediction while emphasising the importance of interpreting predictive performance alongside model behaviour.

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Appendix A: Source Code and Implementation

The complete implementation developed for this study, including data preprocessing, exploratory data analysis, model development, model evaluation, SHAP analyses, and the supplementary evaluation, is available in the following GitHub repository.

GitHub Repository

<https://github.com/NeginHz/heart-disease-xai-thesis>

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